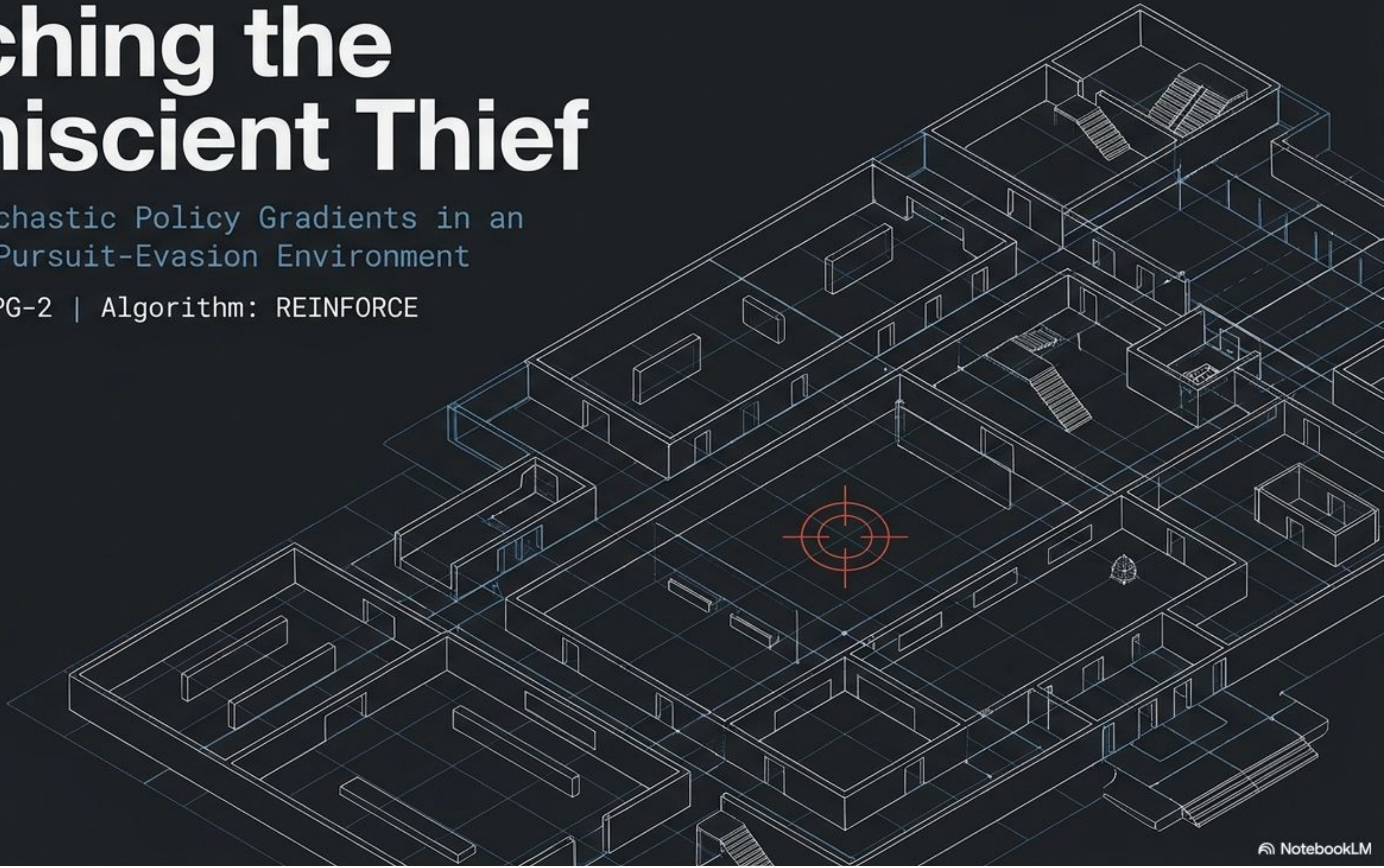
Catching the Omniscient Thief

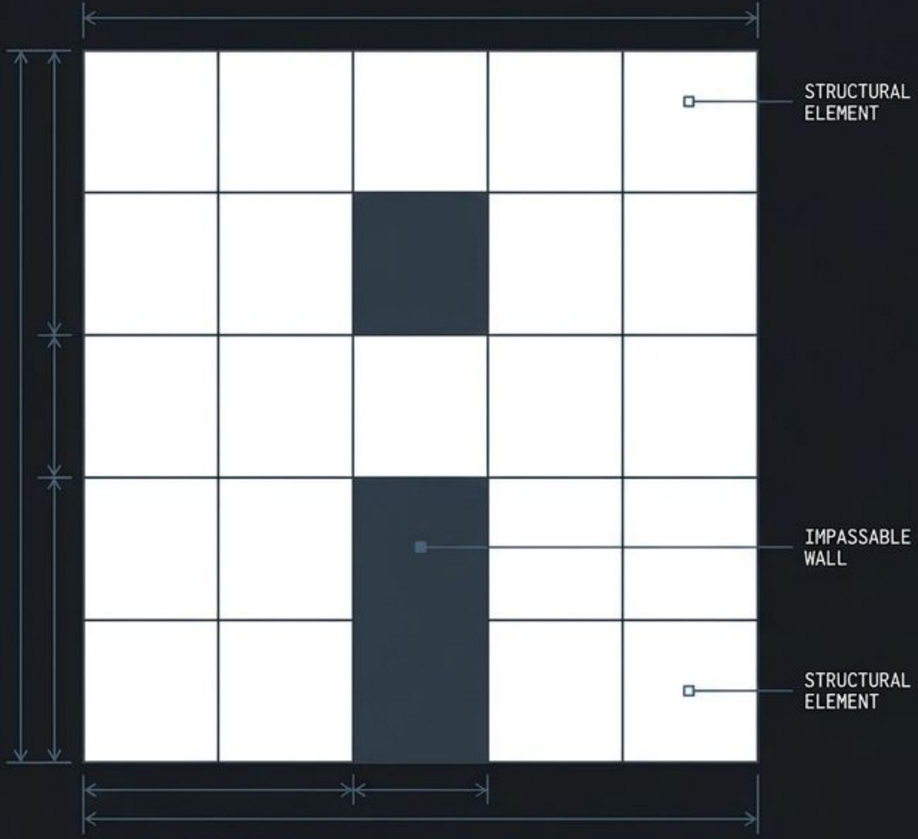
Applied Stochastic Policy Gradients in an Asymmetric Pursuit-Evasion Environment

Project ID: PG-2 | Algorithm: REINFORCE



THE MUSEUM GRIDWORLD

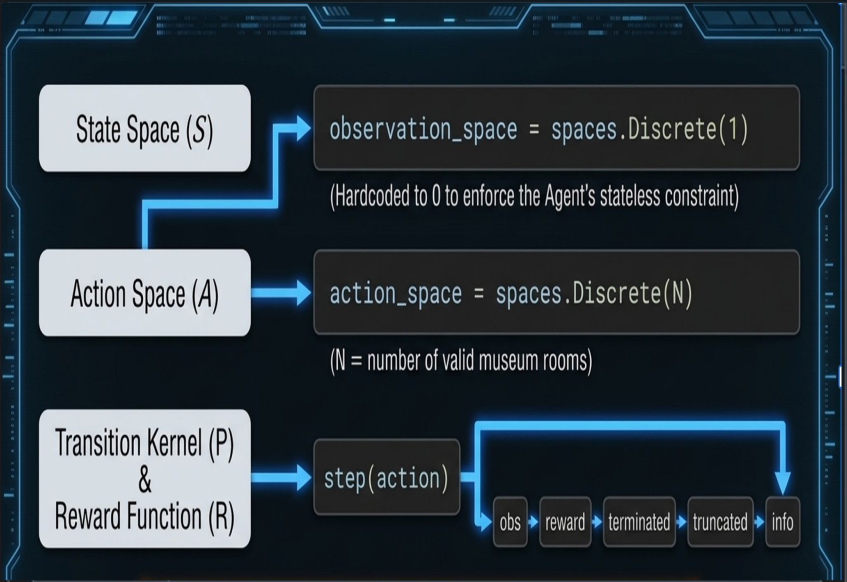
MUSEUM GRIDWORLD SCHEMATIC



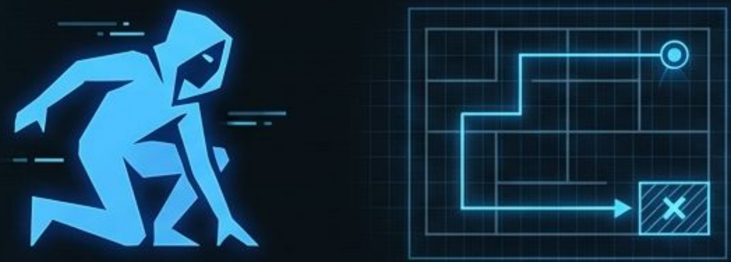
GRID SIZE: N×M (5×5)

THE ARCHITECTURE

Modeled via Gymnasium as a discrete N × M grid. Rooms act as graph vertices connected to a maximum of 4 neighbors via doors.



THE ADVERSARY (THIEF)



- **KNOWLEDGE:** Perfect map awareness. Knows target painting and exit.

- **LOGIC:** Stateful. Computes dynamic shortest-path calculations.

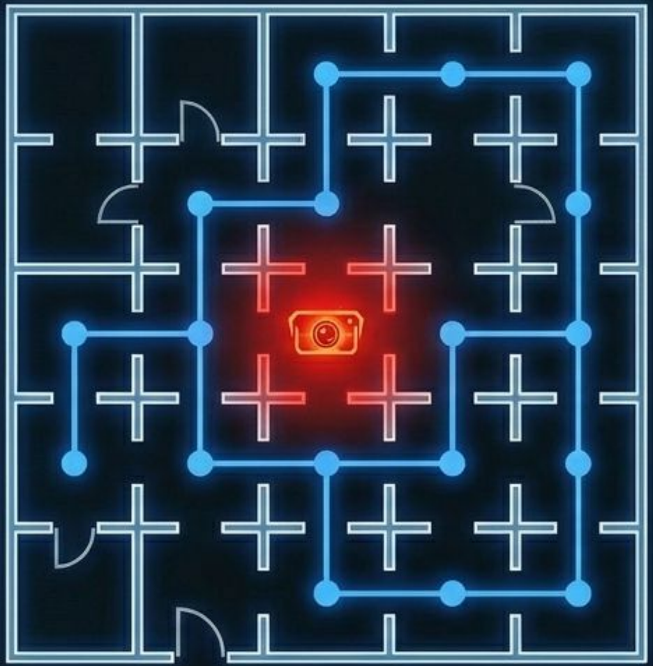
- **INTEL:** Radio contact with a Hacker (knows current camera location).

THE AGENT (SECURITY CAMERA)

- **KNOWLEDGE:** Completely blind to the map, the painting, and the thief.

- **LOGIC:** Stateless. Can only check one room per round.

- **INTEL:** None. Relies purely on delayed reward signals (+1 catch, -1 escape).



$$C(v) = 1 + \beta \cdot 2^{-(n_v - 1)}$$

Cost for the Thief to enter room v.

The Prudence constant (scales aversion to recently checked rooms).

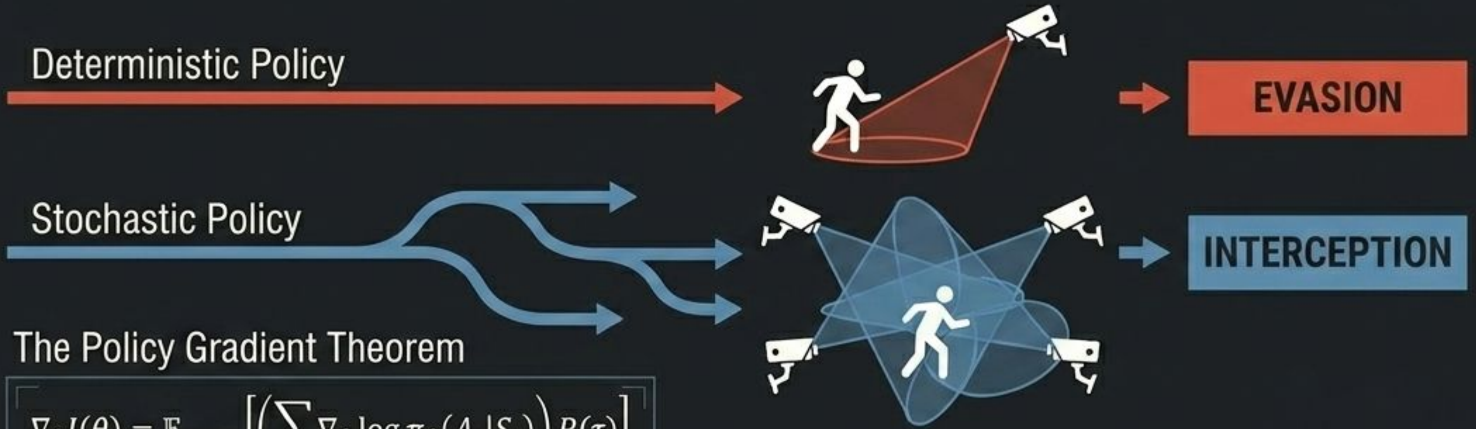
Rounds since room v was last surveilled.

Strategic Takeaway: The camera acts as a temporary, impassable wall. The thief will stall in a safe room rather than blindly walk into active surveillance.

THE THEORETICAL ENGINE: WHY DETERMINISM FAILS

The Flaw of Value-Based RL

Traditional Q-learning maps action-values to states. Here, there are no states.
A deterministic policy creates permanent, exploitable blind spots.



The Policy Gradient Theorem

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta}\left[\left(\sum \nabla_\theta \log \pi_\theta(A_t|S_t)\right) R(\tau)\right]$$

To survive an asymmetric environment, the agent must be unpredictable.

The Goal
Learn an optimal stationary probability distribution across critical chokepoints.

THE AGENT'S BRAIN: STATELESS SOFTMAX

Parameter Vector

The agent learns a raw parameter vector θ, initialized to zero for all N rooms.

Probability Distribution

$$\pi_\theta(a) = \frac{\exp(\theta_a/\tau)}{\sum \exp(\theta'_{a'}/\tau)}$$

The Role of Temperature (τ)

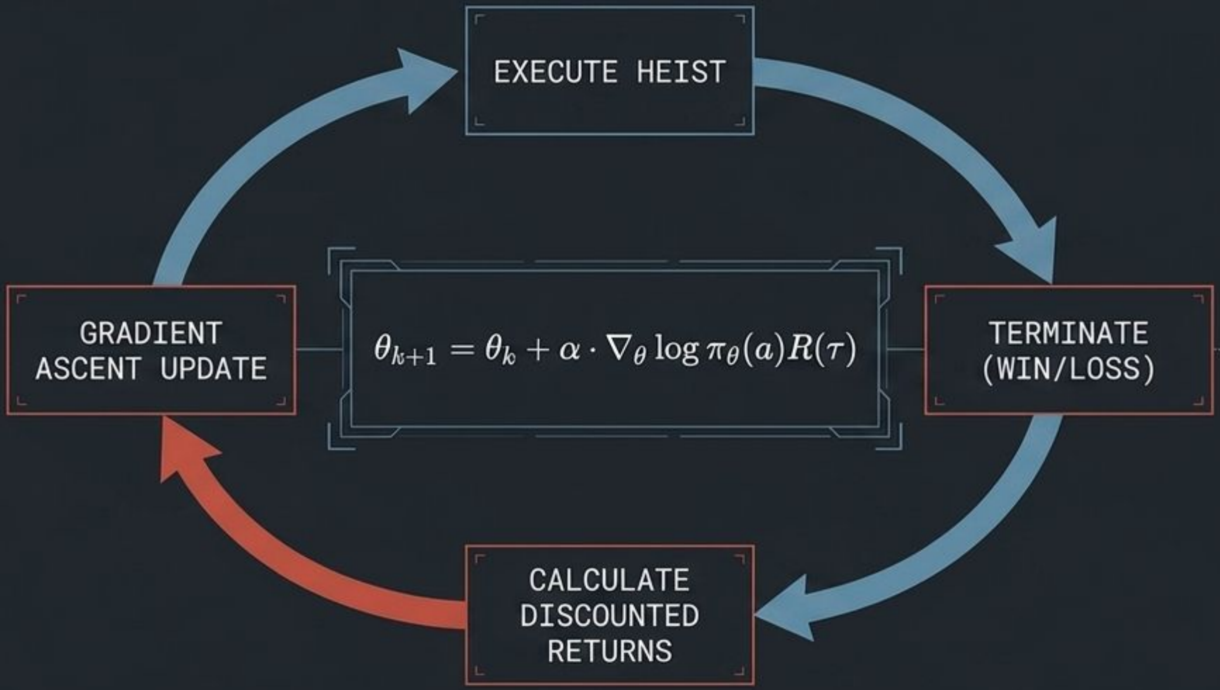
Prevents policy collapse. Stops the agent from permanently fixating on a single room after an early, lucky catch.

Greedy: $\tau = 0.2$
Exploration: $\tau = 20.0$

TRAINING VIA REINFORCE

Monte Carlo Requirement

Requires a full episode trajectory before parameter learning occurs.



$$\theta_{k+1} = \theta_k + \alpha \cdot \nabla_\theta \log \pi_\theta(a) R(\tau)$$

VARIANCE REDUCTION (BASELINES)

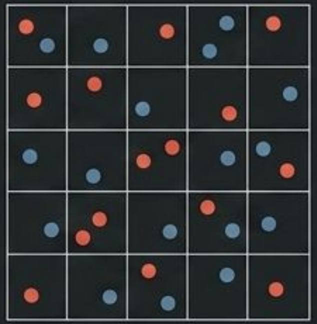
Attempts to center episodic rewards by subtracting the mean return from each step's return, theoretically penalizing below-average actions to stabilize convergence.

THE SPAWN STRATEGY SPECTRUM

Manipulating structural constraints to evaluate the learned probability distribution.

RANDOM

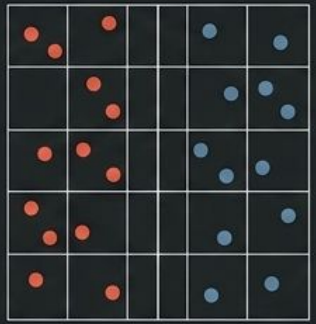
Maximum Chaos



Thief and painting spawn locations are sampled completely at random.

DIVIDED

The Tactical Sweet Spot



Museum split horizontally. Thief spawns exclusively on the left, painting on the right.

FIXED

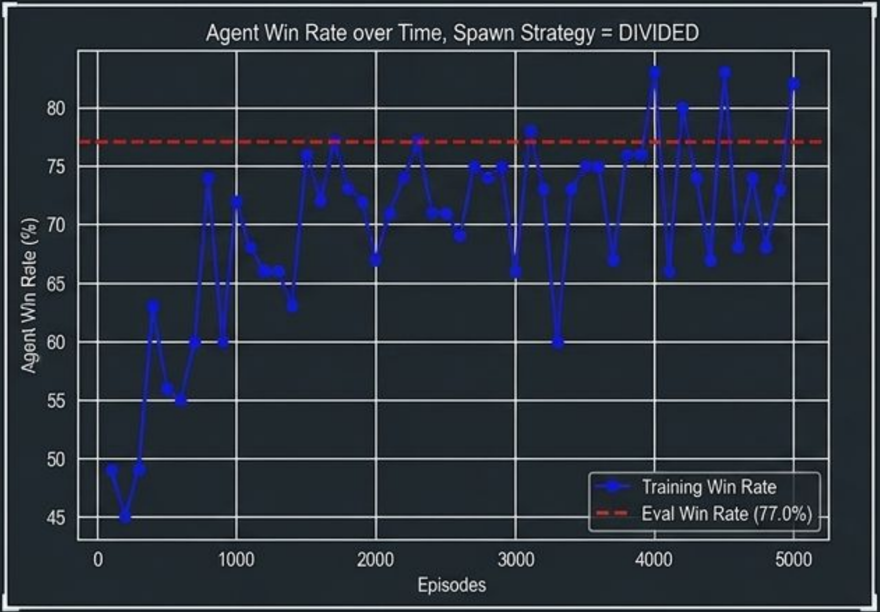
Maximum Predictability

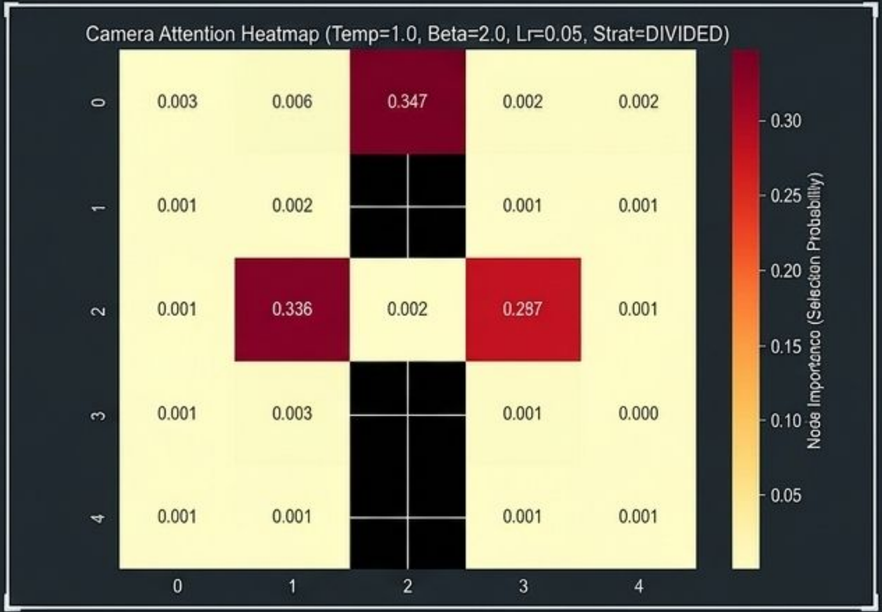
Thief eternally spawns bottom-left, painting eternally top-right.

DATE: 2024.10.27

STATUS: SIMULATION READY

DIVIDED STRATEGY: MASTERING CHOKEPOINTS

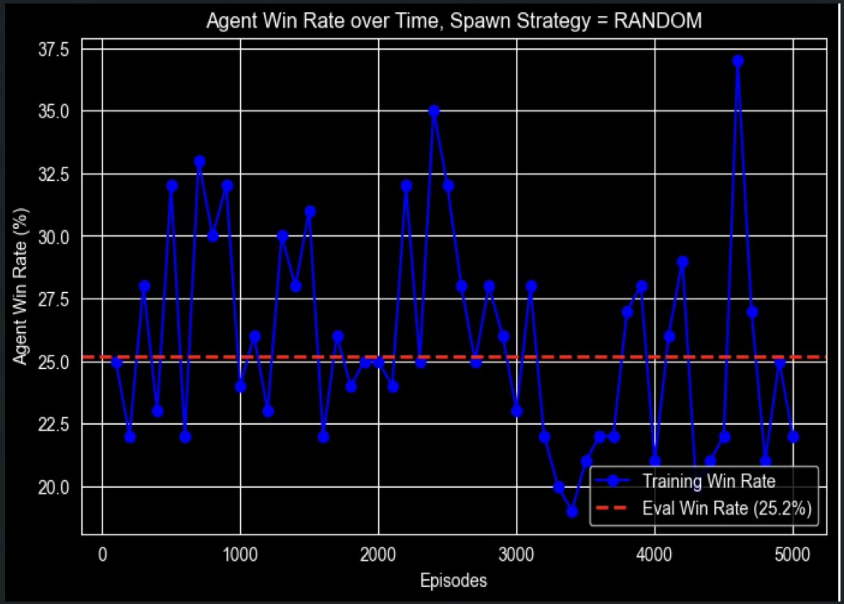


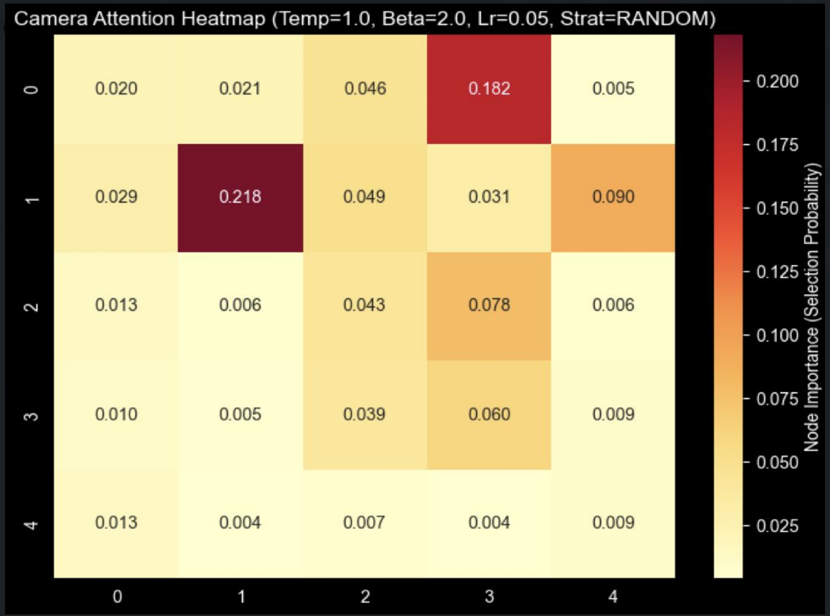


THE SETUP: Thief consistently spawns on the left; the target is on the right.

THE RESULT: Agent achieves a highly stable ~77.0% evaluation win rate (β=2.0, τ=1.0).

RANDOM STRATEGY: ADAPTING TO CHAOS



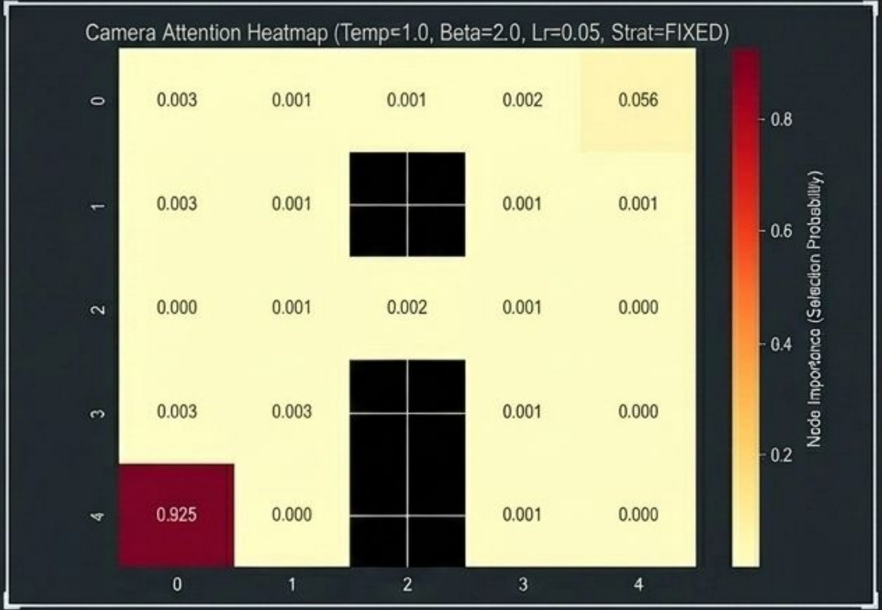


THE SETUP: Start and target positions are sampled entirely at random.

THE RESULT: Evaluation win rate

Drops to ~25%

FIXED STRATEGY: THE ILLUSION OF PERFECTION



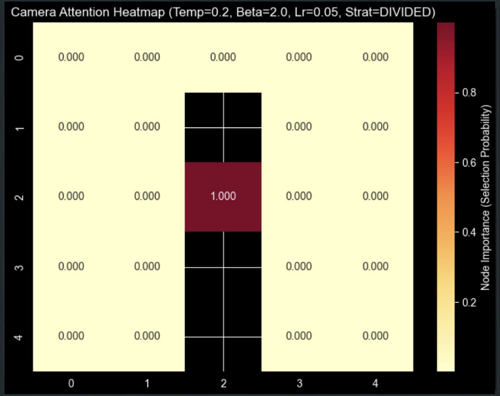
THE SETUP: Thief always begins bottom-left, painting always top-right.

THE RESULT: Learning curve climbs to a near 100% win rate.

TUNING THE ALGORITHM: TEMPERATURE & BASELINES

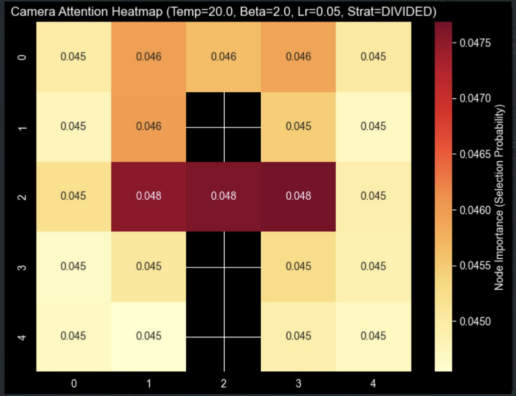
THE TEMPERATURE EXTREMES

τ = 0.2 (GREEDY)



Policy collapses. The agent fixates entirely on a single node after an early catch, guaranteeing evasion on alternate routes.

τ = 20.0 (EXPLORATIVE)



Distribution washes out entirely. Agent fails to capitalize on mathematical bottlenecks.

THE BASELINE PARADOX

Designed to reduce variance by centering returns, applying an average baseline actually degraded performance, pushing the agent to distribute importance too widely.

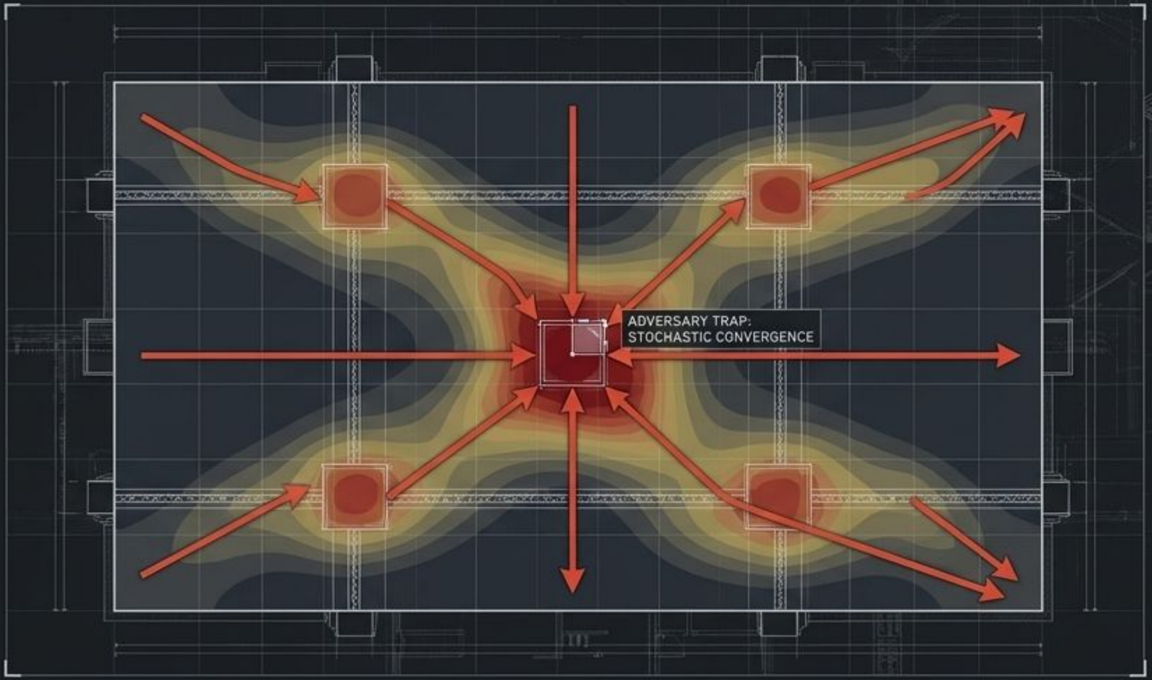
THE NECESSITY OF PROGRAMMED UNPREDICTABILITY

SYNTHESIS

Against an omniscient, state-aware adversary, a blind agent cannot win through deterministic logic. Determinism creates exploitable blind spots.

THE SOLUTION

Stochastic Policy Gradients successfully bridge the asymmetric gap.



By learning geometric chokepoints while maintaining controlled mathematical chaos, the stateless agent effectively traps the omniscient thief.

Thanks

Thanks for listening
Konrad Mocarski